

EEE 101 Electrical Circuits I – v3
EEE 201 Electrical Circuits I - v1, v2

EEE 101L Electrical Circuits I Lab – v3
EEE 202 Electrical Circuits I Laboratory (1.5 credits) – v1, v2

A. Course General Information:

Course Code:	EEE101 EEE101L
Course Title:	Electrical Circuits I Electrical Circuits I Laboratory
Credit Hours (Theory + Laboratory):	3 + 1
Contact Hours (Theory + Laboratory):	3 + 3
Category:	Program Core
Type:	Required, Engineering, Lecture + Laboratory
Prerequisites:	PHY 111 Principles of Physics MAT 110 Mathematics I Differential Calculus and Co-ordinate Geometry
Co-requisites:	None
Equivalent Course	ECE 101 Electrical Circuits I – v3 ECE 201 Electrical Circuits I - v1, v2 ECE 101L Electrical Circuits I Laboratory – v3 ECE 202 Electrical Circuits I Laboratory (1.5 credits) – v1, v2

B. Course Catalog Description (Content):

The course is designed to acquaint students with basic DC electrical circuits and their working. The Kirchhoff's laws, node voltage methodology and circuit theorems are used to solve simple DC circuits' problems. The course then covered the network elements, types of networks & analysis of complex circuits using Mesh current & Nodal voltage method, various circuit theorems such as: Norton's Theorem, Thevenin's Theorem, Superposition Theorem and develop an understanding of how to apply these circuit theorems/techniques for solving different types of complex DC circuit problems having dependent and independent voltage and current sources, ability to apply delta-wye conversion techniques to analyze different types of more complex circuits and calculate maximum power transfer for these circuits. The response of first order RC and RL circuits is also analyzed along with step response. Similar to electric circuit, magnetic circuit also analyzed using basic equations and methods to solve magnetic circuit problems. In addition to class lectures, comprehensive mandatory laboratory (EEE101L) exercises are also designed so that theoretical knowledge may be coincided with practical.

C. Course Objective:

This course is considered as the backbone to fundamental electrical circuits and analysis. Ability to use the techniques, skills and modern engineering tools necessary for modern engineering practice related to DC circuit applications. The rationale of the course is to enable the students to develop the sound understanding of and ability to design and analyze basic electrical DC circuits. As one of the core courses for the EEE program, the knowledge from the course will be applied in future EEE courses such as AC circuits, Electronic devices. The course allows the use the students to use modern engineering techniques, skills and tools to fulfill practical engineering problems related to DC circuit.

D. Course Outcomes (COs):

Upon successful completion of this course, students will be able to

Sl.	CO Description
CO1	Explain the fundamental concepts of linear electrical circuit elements and magnetic properties of materials
CO2	Apply different circuit analysis techniques and circuit theorems to solve circuits for unknown quantities
CO3	Interpret the natural and transient responses of the first order electric systems involving capacitors and inductors
CO4	Use simulation tool to construct DC circuit in schematic level
CO5	Demonstrate basic proficiency in building, debugging and testing basic electrical circuits

E. Mapping of CO-PO-Taxonomy Domain & Level- Delivery-Assessment Tool:

Sl.	CO Description	POs	Bloom's taxonomy domain/level	Delivery methods and activities	Assessment tools
EEE 101 Electrical Circuits					
CO1	Explain the fundamental concepts of linear electrical circuit elements and magnetic properties of materials	a	Cognitive/ Understand	Lectures, notes	Assignment, Quiz
CO2	Apply different circuit analysis techniques and circuit theorems to solve circuits for unknown quantities	a	Cognitive/ Apply	Lectures, notes	Assignment, Quiz, Exam
CO3	Interpret the natural and transient responses of the first order electric systems involving capacitors and inductors	a	Cognitive/ Apply	Lectures, notes	Assignment, Quiz, Exam
EEE 101L Electrical Circuits Lab					
CO4	Use simulation tool to construct DC circuit in schematic level	e	Cognitive/ Understand, Psychomotor/ Manipulation	Lab Class	Lab Work, Lab Exam
CO5	Demonstrate basic proficiency in building, debugging and testing basic electrical circuits	e	Cognitive/ Understand, Psychomotor/ Manipulation	Lab Class	Lab Work, Lab Exam

Mapping of CO - PO Indicators - Knowledge Profile

CO No.	PO No.	Applicable PO Indicator	Knowledge Profile
CO1	a	a1, a3	K1, K3
CO2	a	a4	K3
CO3	a	a4	K3
CO4	e	e3	K6
CO5	e	e1, e3	K6

Description of PO Indicators

PO Description	Indicator Description
(a) Engineering knowledge: Apply knowledge of mathematics, natural science, (<i>electrical and electronic / electronic and communication</i>) engineering fundamentals, and specialization to the solution of complex (<i>electrical and electronic / electronic and communication</i>) engineering problems. (K1 to K4)	a1. Explain the concepts of mathematics, science and engineering knowledge required to solve complex engineering problems
	a3. Apply the appropriate knowledge of natural sciences to solve electrical and electronic engineering problems
	a4. Apply fundamental and specialized engineering knowledge to solve complex electrical and electronic engineering problems
(e) Modern tool usage: Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex (<i>electrical and electronic / electronic and communication</i>) engineering problems, with an understanding of the limitations. (K6).	e1. Select appropriate techniques, resources, modern engineering and IT tools to solve complex engineering problems e3. Use appropriate modern engineering, IT including modelling and simulation tools and techniques to develop and evaluate the solution of complex engineering problems

Description of Applicable Knowledge Profiles

K1	Natural Sciences	A systematic, theory-based understanding of the natural sciences applicable to the discipline
K3	Engineering fundamentals	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
K6	Engineering practice	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline

F. Course Materials:

i. Text and Reference Books:

Sl .	Title	Author(s)	Publication Year	Edition	Publisher	ISBN
1	Fundamentals of Electric Circuits	Charles K. Alexander, Matthew N.O. Sadiku	2009	4 th Ed.	McGraw-Hill	978-0-07-352955-4
2	Introductory Circuit Analysis	Robert L. Boylestad	2015	11 th Ed.	Prentice-Hall	0-13-173044-4

ii. Other materials (if any)

- Lecture notes/presentation slides
- Lab hand-outs

- c. Lab usage manual
- d. Simulation tool (PSpice)

G. Lesson Plan (Theory):

No	Topic	Week/Lecture#	Related CO (if any)
1	Basic definitions and units of electrical parameters, Active and passive circuit elements, Independent and dependent voltage and current sources.	Lecture 1-2	CO1
2	Ohm's law, Kirchhoff's current law, Kirchhoff's voltage law, Series Circuits, Parallel Circuits. Series-parallel circuits, Calculation of equivalent resistance, Voltage and Current division.	Lecture 3-4	CO1
3	Nodal Analysis of electrical circuits using independent voltage/current sources, Nodal Analysis of electrical circuits using dependent voltage/current sources.	Lecture 5-6	CO2
4	Mesh Analysis of electrical circuits using independent voltage/current sources, Mesh Analysis of electrical circuits using dependent voltage/ current sources.	Lecture 7-8	CO2
5	Superposition theorem, Thevenin's theorem using both dependent and independent sources, Norton's theorem, Source transformation theorem.	Lecture 9-12	CO2
Midterm Exam			
6	Delta-Wye transformation, Condition for maximum power transfer calculations.	Lecture 14-16	CO2
7	Inductors and capacitors, Series parallel combination circuits of inductors and capacitors. Responses of RL and RC circuits: Natural responses & Complete responses.	Lecture 17-20	CO3
8	Magnetic fields, Flux density, Permeability, Reluctance, Ohm's Law for magnetic circuits. Magnetizing force, Hysteresis, Ampere's Circuital Law, Determination of flux and magneto motive force. Solving different magnetic circuit problems.	Lecture 21-23	CO1
9	Discussing and solving problems about various topics	Lecture 24	CO2, CO3, CO1
Final Exam			

H. Lesson Plan (Laboratory):

Activity	Week#	Related CO (if any)
Group Formation, Marks Distribution, Report Structure, Project Requirement, Activity List, Course Outcome Assessment, Student Feedback, Google Classroom etc.	1	
Experiment 1 – Verification of KVL and KCL	2	
Experiment 2 – Introduction to Series-Parallel Circuits	3	
Experiment 3 – Verification of Superposition Principle	4	
Experiment 4 – Introduction to PSpice, Introduction to Series-Parallel Circuits and Verification of Superposition Principle using PSpice	5	
Experiment 5 – Verification of Thevenin's Theorem and Maximum Power Transfer Theorem	6	
Recess for Midterm Examination of Theory Courses	7	
Experiment 6 – Introduction to AC Voltage Source and Oscilloscope	8	
Experiment 7 – Study of Transient Behavior of RC Circuit	9	

Experiment 8 – Verification of Thevenin's Theorem and Maximum Power Transfer Theorem and Study of Transient Behavior of RC Circuit using PSpice	10	
Hardware Examination	11	
Project Submission	12	CO4

I. Teaching and Learning Strategy:

Teaching activities primarily rely on the teacher-directed teaching method where the students will receive instructions on the concepts related to the course in sequential manner. Therefore, it's essential that students be present and take notes on all the lectures.

The learning process consists of attending short tutorial sessions, regular tutorial sessions, and quiz/midterm exam script-checking sessions.

Short Tutorial Sessions: These sessions will be held in a classroom setting, where students will be given a problem to solve after the related concepts have been adequately addressed. Students are expected to actively engage in solving the problem to assess their own understanding, with an emphasis on identifying gaps in their knowledge. They are strongly encouraged to consult with one another in their efforts to find a solution.

Regular Tutorial Sessions: These sessions will be organized by the Teaching Assistant and will be conducted in person on campus. During these sessions, students will actively engage in solving problems related to various concepts. If students encounter difficulties, they are expected to seek one-on-one explanations. Peer-to-peer assistance is strongly encouraged.

Teaching Assistants are strictly instructed not to solve any problems on the whiteboard upon request unless students have demonstrated active participation in the problem-solving process. Additionally, TAs will not provide consultations on conceptual topics, as separate consultation sessions are allocated for that purpose.

This exercise aims to help students identify gaps in their understanding, address those gaps, and ultimately solidify their grasp of key concepts.

Script Checking Sessions: These sessions aim to provide feedback on different assessment activities. During script-checking sessions (SC sessions), students must bring their class notes. Feedback on exam mistakes will be provided, serving as an essential tool for enhancing topic comprehension. This approach is based on the principle that identifying mistakes highlights gaps in understanding and indicates what not to do. Recognizing these gaps provides a starting point for corrective measures to strengthen learning.

Student participation in Regular tutorial sessions and Script checking sessions will be recorded and considered a component of their overall evaluation.

J. Assessment Tools (Theory):

Assessment Tools	Weightage (%)
Attendance	5
Assignment	10
Quizzes**	30
Midterm Exam	20

Final Exam	35
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**** (N-1) Rules to be applied in selecting quizzes. N=Number of quizzes.**

J. Assessment Tools (Laboratory):

Assessment Tools	Weightage (%)
Attendance	10
Hardware class performance	10
Hardware report	10
Software report	10
Project	30
Hardware examination	30

K. CO Assessment Plan:

Assessment Tools	Course Outcomes				
	CO1	CO2	CO3	CO4	CO5
Quizzes (Q)	√	√	√		
Assignment		√	√		
Midterm exam	√	√			
Final Exam		√	√		
Lab work				√	√
Lab Exam.				√	√

L. CO Attainment Policy:

As per Department of EEE Course Outcome Attainment Policy

M. Grading policy:

As per Brac University grading policy

N. Course Instructor (For Section 1): Dr. Belal Bhuian

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